

МИНИСТЕРСТВО ОБРАЗОВАНИЯ РЕСПУБЛИКИ БЕЛАРУСЬ
БЕЛОРУССКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
Факультет прикладной математики и информатики
Кафедра информационных систем управления

КУРЕЦ ЛЮБОВЬ ОЛЕГОВНА
Отчет по Лабораторной работе №2

Составление отчетов. Транспонирование
резльтирующих множеств
студентки 3 курса 2 группы

Преподаватель
Кузьмина А.В.
Доцент кафедры ИСУ

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Задание 1

```
select * from (
  select JOBNAME from CAREER11 c left join JOB11 j on c.JOBNO = j.JOBNO
)
pivot(
  count(JOBNAME) for JOBNAME in (
    'PRESIDENT' as PRESIDENT,
    'FINANCIAL DIRECTOR' as FINANCIAL_DIRECTOR,
    'EXECUTIVE DIRECTOR' as EXECUTIVE_DIRECTOR,
    'MANAGER' as MANAGER,
    'CLERK' as CLERK,
    'DRIVER' as DRIVER,
    'SALESMAN' as SALESMAN)
);
```

```
select * from (  select JOBNAME from CAREER11 c left join JOB11 j on c.JOBNO = j.JOBNO ) pivot(
count(JOBNAME) for JOBNAME in ( 'PRESIDENT' as PRESIDENT, 'FINANCIAL DIRECTOR' as FINANCIAL_DIRECTOR,
'EXECUTIVE DIRECTOR' as EXECUTIVE_DIRECTOR, 'MANAGER' as MANAGER, 'CLERK' as CLERK, 'DRIVER' as
DRIVER, 'SALESMAN' as SALESMAN) )
```

PRESIDENT	FINANCIAL_DIRECTOR	EXECUTIVE_DIRECTOR	MANAGER	CLERK	DRIVER	SALESMAN
1	3	1	0	3	1	2

Задание 2

```
select
max(case when DEPTNAME = 'ACCOUNTING' then EMPNAME else NULL end) as
ACCOUNTING,
max(case when DEPTNAME = 'RESEARCH' then EMPNAME else NULL end) as
RESEARCH,
max(case when DEPTNAME = 'SALES' then EMPNAME else NULL end) as SALES,
max(case when DEPTNAME = 'OPERATIONS' then EMPNAME else NULL end) as
OPERATIONS
from (
select DEPTNAME, EMPNAME, row_number()over(partition by DEPTNAME order
by EMPNAME) rn
from EMP11 join (CAREER11 join DEPT11 using (DEPTNO)) using (EMPNO)
)
```

group by rn;

```
select max(case when DEPTNAME = 'ACCOUNTING' then EMPNAME else NULL end) as ACCOUNTING, max(case when DEPTNAME = 'RESEARCH' then EMPNAME else NULL end) as RESEARCH, max(case when DEPTNAME = 'SALES' then EMPNAME else NULL end) as SALES, max(case when DEPTNAME = 'OPERATIONS' then EMPNAME else NULL end) as OPERATIONS from ( select DEPTNAME, EMPNAME, row_number()over(partition by DEPTNAME order by EMPNAME) rn from EMP11 join (CAREER11 join DEPT11 using (DEPTNO)) using (EMPNO) ) group by rn
```

ACCOUNTING	RESEARCH	SALES	OPERATIONS
ALLEN	JOHN MARTIN	ALLEN	ALEX BOUSH
RICHARD MARTIN	SMITH	ALLEN	JOHN KLINTON
RICHARD MARTIN	-	SMITH	-

Задание 3

```
select j.JOBNAME,  
case j.JOBNAME  
when 'CLERK' then emp_cnts.CLERK  
when 'DRIVER' then emp_cnts.DRIVER  
when 'EXECUTIVE DIRECTOR' then emp_cnts.EXECUTIVE_DIRECTOR  
when 'FINANCIAL DIRECTOR' then emp_cnts.FINANCIAL_DIRECTOR  
when 'MANAGER' then emp_cnts.MANAGER  
when 'PRESIDENT' then emp_cnts.PRESIDENT  
when 'SALESMAN' then emp_cnts.SALESMAN  
else NULL  
end as EMPCOUNT  
from (  
select * from (  
select JOBNAME from CAREER11 c left join JOB11 j on c.JOBNO = j.JOBNO  
)  
pivot(  
count(JOBNAME) for JOBNAME in (  
'PRESIDENT' as PRESIDENT,  
'EXECUTIVE DIRECTOR' as EXECUTIVE_DIRECTOR,  
'FINANCIAL DIRECTOR' as FINANCIAL_DIRECTOR,  
'MANAGER' as MANAGER,  
'CLERK' as CLERK,  
'DRIVER' as DRIVER,  
'SALESMAN' as SALESMAN)  
)  
) emp_cnts, (  

```

```
select JOBNAME from JOB11) j;
```

```
select j.JOBNAME, case j.JOBNAME when 'CLERK' then emp_cnts.CLERK when 'DRIVER' then emp_cnts.DRIVER
when 'EXECUTIVE DIRECTOR' then emp_cnts.EXECUTIVE_DIRECTOR when 'FINANCIAL DIRECTOR' then
emp_cnts.FINANCIAL_DIRECTOR when 'MANAGER' then emp_cnts.MANAGER when 'PRESIDENT' then
emp_cnts.PRESIDENT when 'SALESMAN' then emp_cnts.SALESMAN else NULL end as EMPCOUNT from ( select *
from ( select JOBNAME from CAREER11 c left join JOB11 j on c.JOBNO = j.JOBNO ) pivot(
count(JOBNAME) for JOBNAME in ( 'PRESIDENT' as PRESIDENT, 'EXECUTIVE DIRECTOR' as EXECUTIVE_DIRECTOR,
'FINANCIAL DIRECTOR' as FINANCIAL_DIRECTOR, 'MANAGER' as MANAGER, 'CLERK' as CLERK, 'DRIVER' as
DRIVER, 'SALESMAN' as SALESMAN) ) ) emp_cnts, ( select JOBNAME from JOB11) j
```

JOBNAME	EMPCOUNT
MANAGER	0
FINANCIAL DIRECTOR	3
EXECUTIVE DIRECTOR	1
SALESMAN	2
CLERK	3
DRIVER	1
PRESIDENT	1

Задание 4

Создание таблицы:

```
create or replace view EMPS_INFO11 as (
select JOBNO, JOBNAME, count(JOBNAME) as JOBCOUNT
from CAREER11 join JOB11 using(JOBNO)
group by JOBNAME, JOBNO
);
```

Запрос:

```
select
case rn
when 1 THEN JOBNAME
when 2 THEN cast(JOBCOUNT as CHAR(4))
end EMPS
from (
select e.JOBNAME, e.JOBCOUNT,
row_number()over(partition by e.JOBNAME order by e.JOBNAME) rn
from EMPS_INFO11 e,
(select * from EMPS_INFO11 where ROWNUM <= 3)
);
```

```
select case rn when 1 THEN JOBNAME when 2 THEN cast(JOBCOUNT as CHAR(4)) end EMPS from ( select
e.JOBNAME, e.JOBCOUNT, row_number()over(partition by e.JOBNAME order by e.JOBNAME) rn from EMPS_INFO11
e, (select * from EMPS_INFO11 where ROWNUM <= 3) )
```

EMPS
CLERK
3
-
DRIVER
1
-
EXECUTIVE DIRECTOR
1
-
FINANCIAL DIRECTOR
3
-
PRESIDENT
1
-
SALESMAN
2
-

Задание 5

```
select decode(lag(DEPTNAME) over(order by DEPTNAME), DEPTNAME, NULL,
DEPTNAME) DEPTNAME, EMPNAME
from EMP11 join (CAREER11 join DEPT11 using(DEPTNO)) using(EMPNO);
```

```
select decode(lag(DEPTNAME) over(order by DEPTNAME), DEPTNAME, NULL, DEPTNAME) DEPTNAME, EMPNAME from EMP11 join (CAREER11 join DEPT11 using(DEPTNO)) using(EMPNO)
```

DEPTNAME	EMPNAME
ACCOUNTING	ALLEN
-	RICHARD MARTIN
-	RICHARD MARTIN
OPERATIONS	ALEX BOUSH
-	JOHN KLINTON
RESEARCH	SMITH
-	JOHN MARTIN
SALES	ALLEN
-	ALLEN
-	SMITH